

# Assignment 1

## Computational Fracture Mechanics (WiSe2526)

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### Given Data

In this section, the parameters value for the material properties, that required by the author to calculate it based on the author's Immatriculation Nummer, will be defined accordingly. To make the calculation detailed and transparent, each parameter will be derived step by step.

$$d_1 = 4, \quad d_2 = 6, \quad d_3 = 8 \quad (1)$$

#### Parameter for 1st question

- Plate length  $b$   
 $b = 2.0 + 0.1d_1 = 2.0 + 0.1 \times 4 = 2.4 \text{ mm}$
- Plate width  $h$   
 $h = b(0.4 + 0.04d_2) = 2.4(0.4 + 0.04 \times 6) = 1.536 \text{ mm}$
- Crack length  $l$   
 $l = b(0.05 + 0.02d_3) = 2.4(0.05 + 0.02 \times 8) = 0.504 \text{ mm}$
- Applied displacement  $u$   
 $u = 0.001 + \frac{d_3}{1000} [\text{mm}] = 0.001 + \frac{8}{1000} = 0.009 \text{ mm}$

#### Parameter for 2nd question

- Applied displacement  $u$   $u = 0.1 + \frac{d_3}{100} [\text{mm}] = 0.1 + \frac{8}{100} = 0.18 \text{ mm}$

# 1 Introduction

In this first assignments, the main focus will be on performing crack analysis using finite element method (FEM) in Abaqus software using two approaches. The first method is to perform a pre-defined crack analysis without separation between two materials, whereas the area around the crack tip will be evaluated to obtain the behavior of the stress field. Hence, J-integral will be calculated based on the stress field using domain integral method to obtain the energy release quantity. The key takeaway from this is to find the stable value of J-integral based on the contour integral surrounding the crack, which will be used as a reference value for the crack initiation value.

The second task is to analyze crack propagation using the cohesive zone method. Hence, some cohesive elements will be defined along the pre-defined crack path with the cohesive parameters with an initial crack opening on the front of the cohesive line. From the task definition, the separation behavior of the cohesive are will be observed, as well as the energy release rate. This can be achieved by plotting the traction-separation curve, which will show the relationship between the traction and separation displacement along the cohesive zone. Another key aspect that will be obtained is to know the force-time response during the crack propagation, so that it can be correlated with the traction-separation curve.

## 2 Model Set up

### 2.1 Material Properties

Table 1: Flow curve data for Q1 and Q2

| Plastic Stress (MPa) | Strain |
|----------------------|--------|
| 300                  | 0      |
| 400                  | 0.001  |
| 600                  | 0.003  |
| 700                  | 0.005  |
| Young Modulus (GPa)  | $\nu$  |
| 210                  | 0.3    |

Table 2: Cohesive parameters for Q2

| Material Parameters  | Values |
|----------------------|--------|
| $K_n$ (GPa)          | 210    |
| $K_s$ (GPa)          | 210    |
| $K_t$ (GPa)          | 210    |
| $\sigma_n$ (MPa)     | 500    |
| $\sigma_{s,t}$       | 0      |
| $\delta_{fail}$ (mm) | 0.0038 |

Table 1 shows the material properties for the elastic and plastic part, which includes the plastic flow with its strain as well as the young modulus and Poisson's ratio for the elastic part. Here, the material will be used for both question 1 and question 2. Meanwhile, table 2 shows the cohesive parameters, which will be used to define the cohesive zone model in question 2. The parameters include the stiffness of the cohesive area in normal and shear directions, the maximum nominal stress, and the failure separation displacement.

## 2.2 Geometry and Boundary Conditions

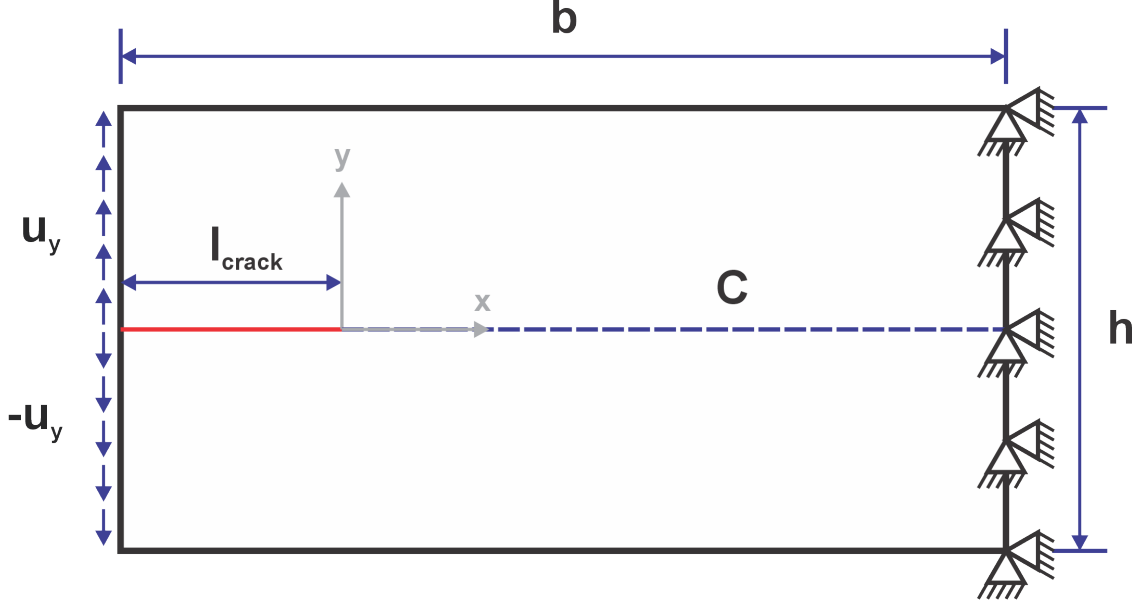


Figure 1: Geometry and boundary definition of the plate with horizontal crack. The geometry parameters are defined in the first section of the report, whereas  $b = 2.4$  mm,  $h = 1.536$  mm, and  $l = 0.504$  mm. This dimension configuration is used for both question 1 and 2.

Figure 1 shows the geometry and boundary conditions of the plate with horizontal crack. Hence, the displacement will be applied on the left side of the plate, whereas the left-top edge will have displacement to the  $y$ -positive direction and left-bottom edge will have the opposite direction. The right side of the plate will be fixed in both  $x$  and  $y$  directions. For the first task, a crack with length of  $l$  is defined on the left-middle side of the plate. This definition will ensure a separation, meaning that the nodes are separated with this crack line.

For the second task, cohesive area is defined along the  $C$  length from the crack tip until the edge of the right side of the plate. To achieve this, two separate geometry are created. This consist of top and bottom plate, which then both of them is connected using contact definition. Hence, the contact is defined as a cohesive contact, where it spans through the  $C$  line according to figure 1. The cohesive parameters are defined as in table 2. Since the cohesive area is defined only in the  $C$  line, therefore the non-contact area (hence is  $l_{crack}$ ) is consider as the crack initiation length.

## 2.3 Mesh Set Up

The next step after setting up the geometry and boundary condition is the mesh definition. Since there is no constraint in terms of element numbers as well as computational power restriction, a fine mesh is used for both questions.

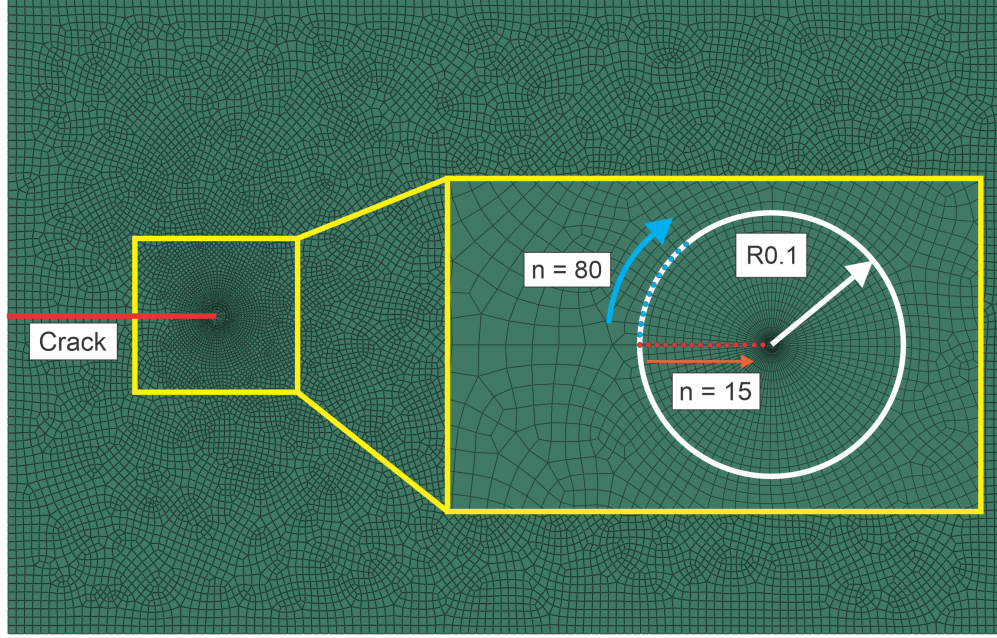


Figure 2: Mesh configuration for the first task. A standard and linear CPS4 (4-node bilinear plane stress quadrilateral) mesh configuration is implemented with the possibility to change to plane stress and plane strain (CPE4) condition, without reduced integration mode. Additional refinement is added to the crack tip with more structured mesh with 15 nodes mesh partitioning.

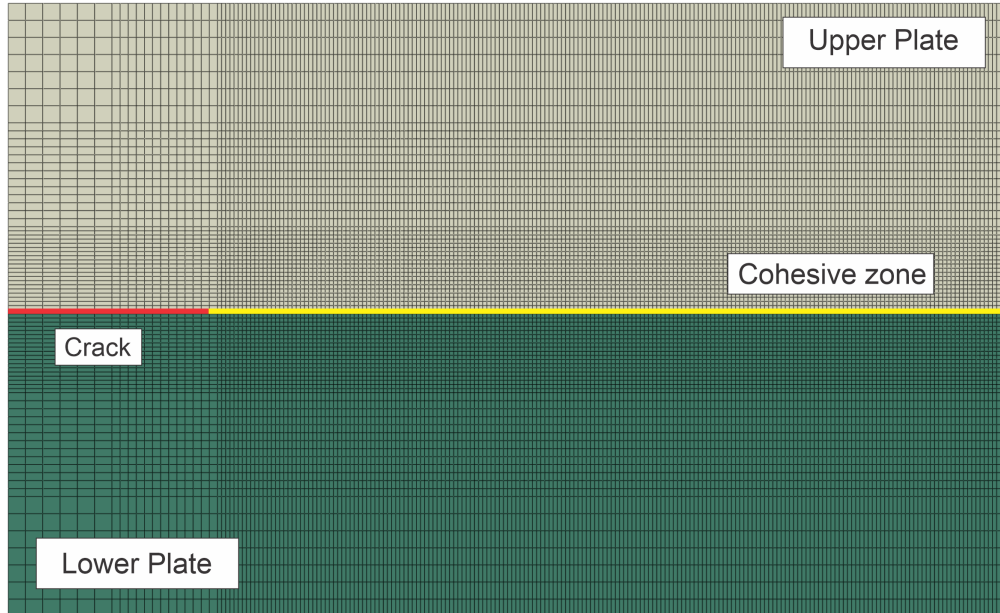


Figure 3: Final mesh configuration for both question 1 and question 2. The mesh is created using CPE4 element type, which is a 4-node bilinear plane strain quadrilateral element. The mesh is refined around the crack tip area to accurately capture the stress concentration effects. More detailed mesh sizing configuration is shown in figure 16.



# Results and Discussion

## Q1: Crack in plate under uniaxial tension

### Elastic Analysis

... Using the elastic material parameters, generate a contour plot of the von Mises stress near the crack tip. Extract and plot the relevant stress components along path  $C$  (see Figure 1) for both plane strain and plane stress conditions. Compare and discuss the differences between the two cases.

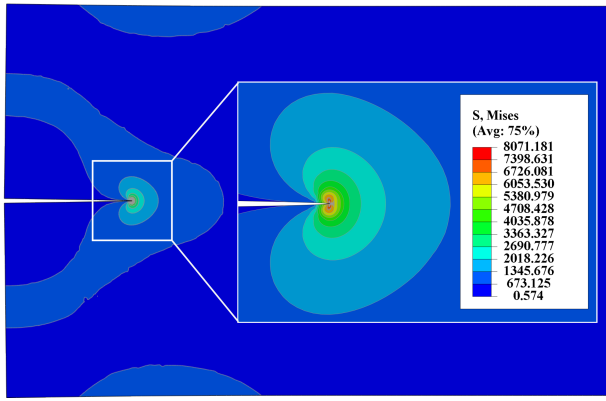


Figure 4: Von Mises stress contour plot around the crack tip for elastic analysis under plane stress condition.

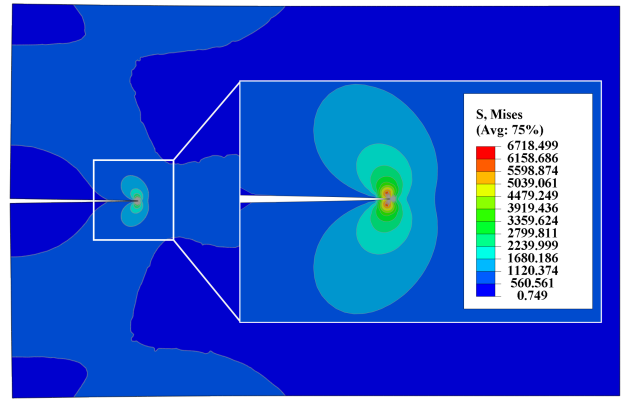


Figure 5: Von Mises stress contour plot around the crack tip for elastic analysis under plane strain condition.

Both figures above shows the stress field for the plane stress and strain condition for figure 4 and 5 respectively. In plane stress, the Von-Misses stress area as well as the value are larger compare to the plane strain. The area shows more rounded shape with an almost equal radius, while the plane strain created a larger stress distribution near the back tip of the crack.

This difference is caused by the different set-up and definition of plane stress and strain model. Where in the plane stress, the out-plane stress is essentially zero:

$$\sigma_z = 0 \quad (2)$$

Whereas in the plane strain, the constraint is located in the strain out-of-plane is defined as follows:

$$\varepsilon_z = 0 \quad (3)$$

Theoretically, as the strain is constraint in the out-plane direction, additional stress will be generated due to the restriction of movement in one plane. However, the main context here is the Von-Misses stress, which basically dependant to the deviatoric part of the stress

instead of the hydrostatic part. Hence, the basic Von-Misses formulation is calculated as follows:

$$\sigma_{vm} = \sqrt{\frac{3}{2}s_{ij}s_{ij}} \quad (4)$$

Where the deviatoric shear component can be described using the hydrostatic term  $\sigma_h$  in following equation:

$$s_{ij} = \sigma_{ij} - \sigma_h \delta_{ij} \quad (5)$$

As mentioned before, the plane strain condition will increase the triaxiality stress component ( $\eta = \frac{\sigma_h}{\sigma_{vm}}$ ), by means of increasing the hydrostatic stress due to restriction of one axis freedom, consequently generating additional stress in that direction. Hence, the shear component consequently decreased as well as the overall Von-Misses stress. While in plane stress, the shear component will be larger compare to plane strain as it allows more freedom in the out-plane movement, which therefore increase the Von-Misses value. To visualize the stress distribution numerically, figure 6 shows the Von-Misses distribution along path C, which describe additionally in the graph picture.

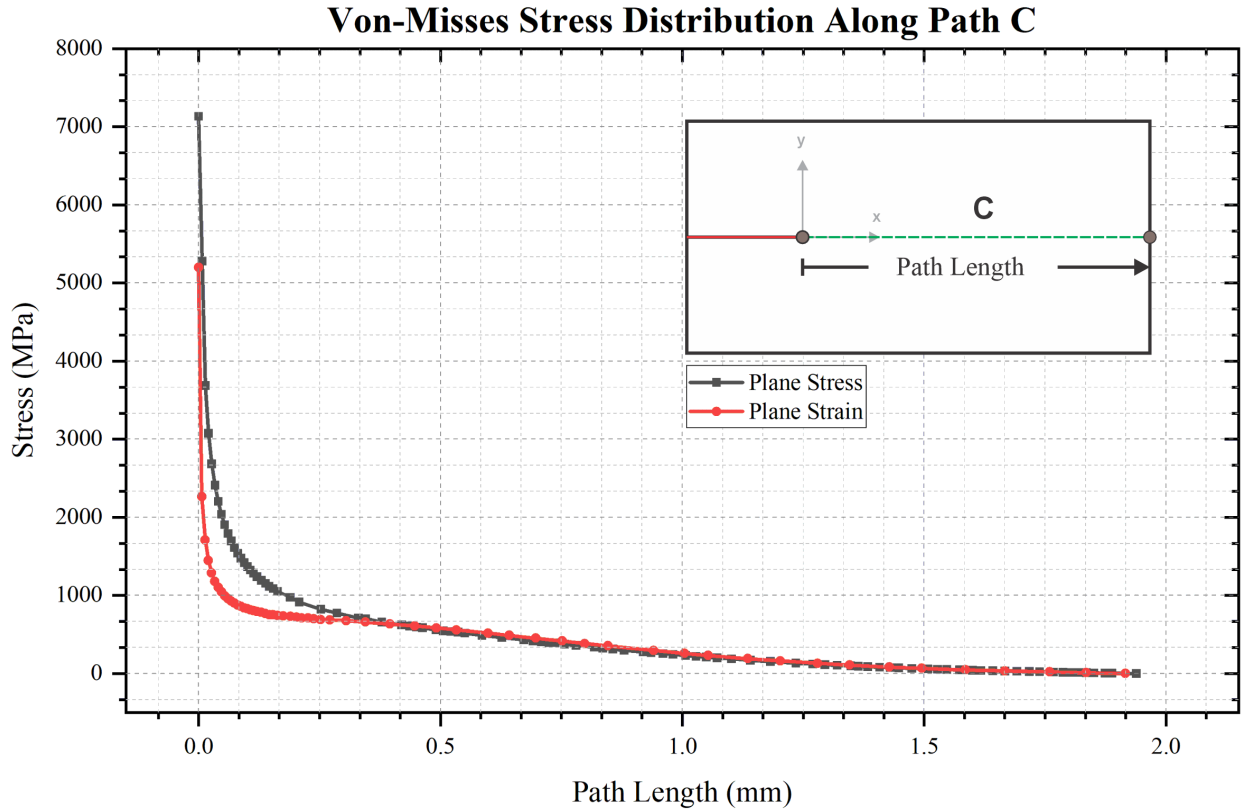


Figure 6: Stress components along path C for both plane stress and plane strain conditions in elastic analysis.

## Elastic-plastic analysis

... Now include plasticity using the material parameters given in Table 1. Produce a contour plot of the von Mises stress near the crack tip. Extract and plot the relevant stress and strain components along path C for both plane strain and plane stress conditions. Discuss:

- The differences between the elastic and elastic-plastic responses
- The differences in the size and shape of the stress field around the crack tip for plane strain vs. plane stress

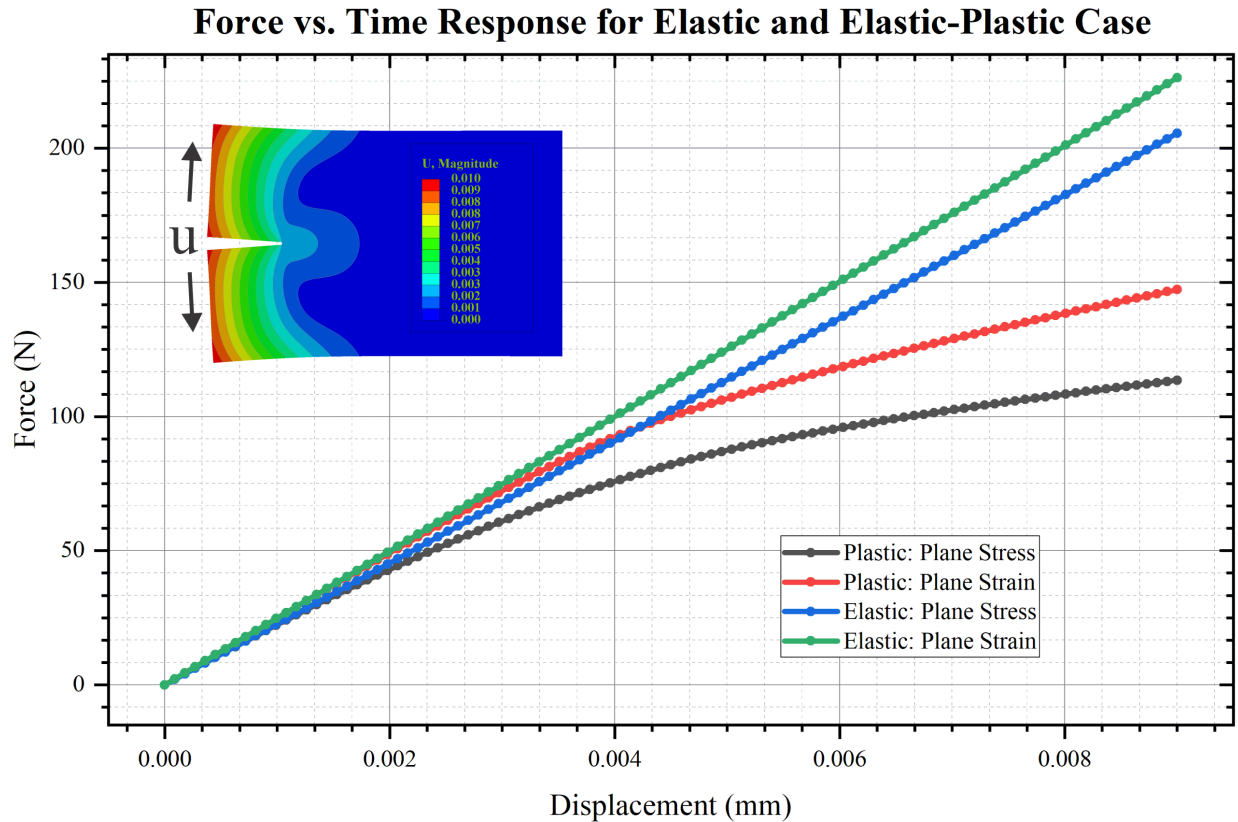


Figure 7: Von Mises stress comparison between elastic and elastic-plastic analysis under plane stress condition.

Figure 7 shows the reaction force plot curve respectively to the displacement, which have been probed in one of the edges of the applied displacement boundary condition. For both elastic and plastic case, the plane strain generated larger forces compare to the plane stress. This can be explained as well with the previous description of plane stress and strain in the first part of Q1, that plane strain created more hydrostatic stress component due to the restriction of one of the out-plane strain component of  $\varepsilon_z$ . Hence, the probed value of reaction force in u direction, which is proportional to the  $\sigma_{22}$ , will also produce larger quantities.

Comparing the elastic and plastic behavior, all curves show a good agreement in the physical interpretation of elastic and plastic material behavior. Whereas in the plastic variation, there are a softening of material due to the defined plasticity after the yielding region. This is showing a contrast in the elastic material, which plasticity is not introduced in the model. Therefore, a continuous linear-behavior is generated throughout a larger displacement.

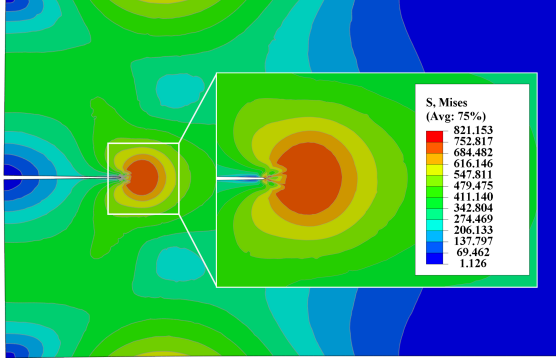


Figure 8: Von Mises stress contour plot around the crack tip for plastic analysis under plane stress condition.

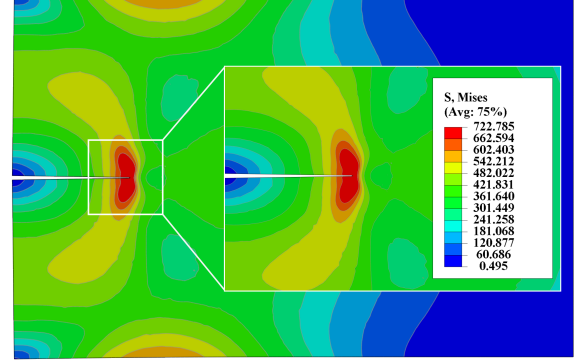


Figure 9: Von Mises stress contour plot around the crack tip for plastic analysis under plane strain condition.

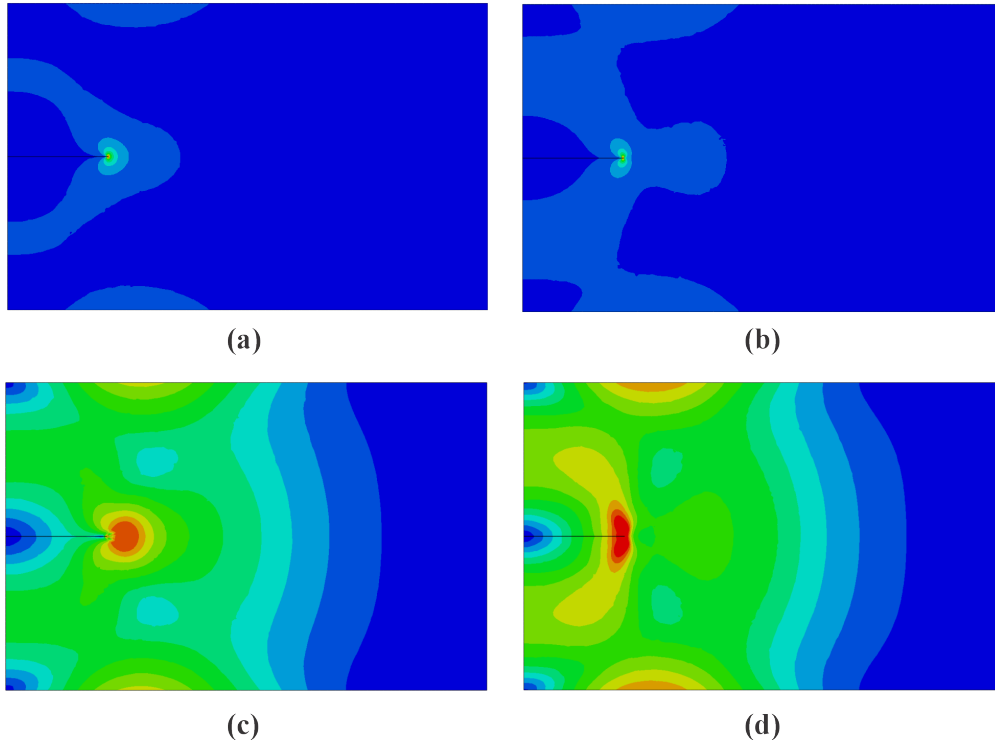


Figure 10: Stress components along path C for both plane stress and plane strain conditions in elastic-plastic analysis. (a) Elastic plane stress, (b) elastic plane strain, (c) plastic plane stress, (d) plastic plane strain

To point out the differences of the stress distribution point, a list of the visualization description is created to point out and highlight the behavior for each case:

- In plastic-plane stress case figure 10(c), the stress concentration area is smaller compared in the plane strain condition shows in figure 10(d).
- The plastic-plane strain produced larger area in the von-Misses in an oval-like shape spreading dominantly in the vertical direction.
- Both elastic-plane stress and strain create significantly smaller area compare to the elastic-plastic case, however the value is significantly larger.

Both figure 8 and 9 show the Von-Misses stress distribution for both plane stress and strain cases, respectively. Compare to the elastic cases, elastic-plastic conditions have smaller value in terms of the Von-Misses stress value. This can be related to figure 7, that the force reaction has larger value in the elastic part compare in the plastic material. Since plasticity is introduced and softening is implemented after the material exceeds its yielding point, the generated stress will have smaller value compare to the elastic material if compared in the same displacement. In the C-curve comparison showed in figure 11, near the cracked tip showed a significant difference of Von-Misses stress value, especially for the elastic and plastic comparison, the differences are quite large for both plane stress and strain cases.

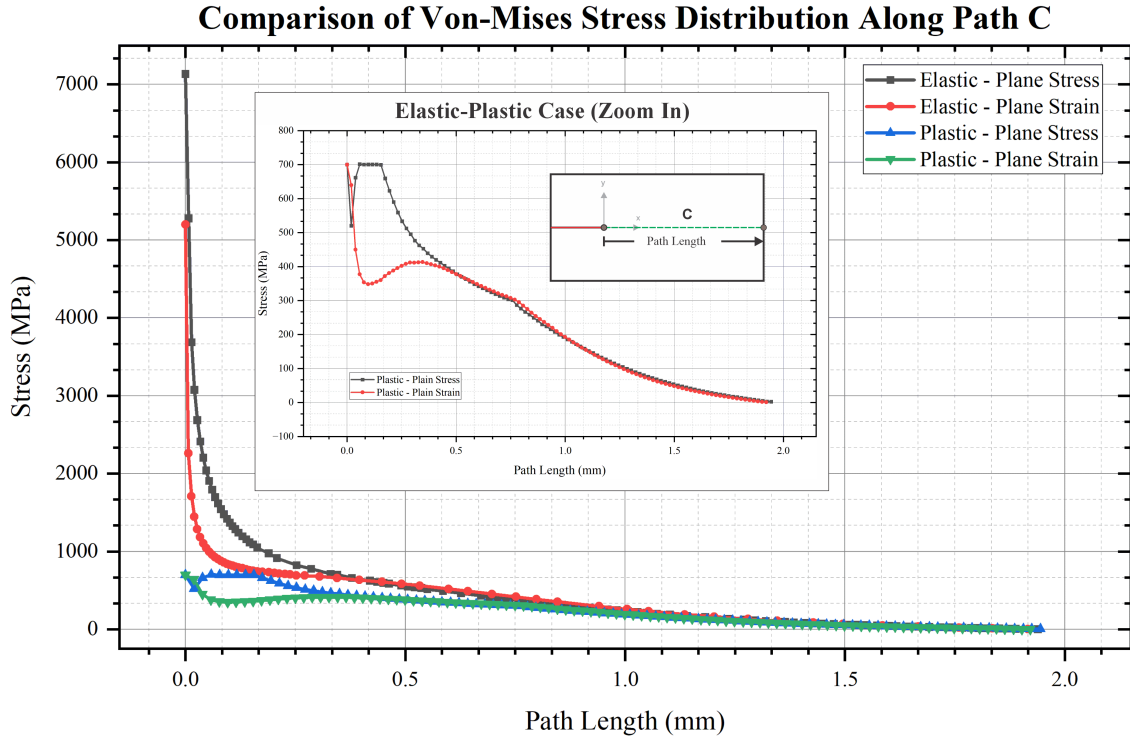


Figure 11: Stress components along path C for both plane stress and plane strain conditions in elastic-plastic analysis.

### J-integral evaluation

... For both elastic and elastic-plastic cases, evaluate the  $J$ -integral over 10 contours around the crack tip only under plane stress condition. Plot the  $J$ -integral value as a function of contour number and use this plot justify which contour range you consider reliable for reporting the final  $J$  value.

- For the purely elastic material model under plane stress condition, compute the  $J$ -integral
- Repeat the  $J$ -integral computation including plasticity. ( $\alpha = 0.01$ ,  $n = 10$ )
- Compare the  $J$ -integral results from pure elastic material model and elastic-plastic model, discuss the differences based on the theoretical  $J$ -integral calculation methods given by the lecture.

In this section,  $J$ -integral values are evaluated for both elastic and elastic-plastic cases in plain stress condition. To define the elastic-plastic behavior, Ramberg-Osgood flow curve is generated to describe the Plasticity region in the flow curve. Ramberg-Osgood relation is defined as following equation

$$\varepsilon = \frac{\sigma}{E} + \alpha \left( \frac{\sigma}{\sigma_y} \right)^n \quad (6)$$

Where  $\varepsilon$  is the total strain,  $\sigma$  is the stress,  $E$  is the Young's modulus,  $\sigma_y$  is the yield stress,  $n$  is the strain hardening exponent, and  $\alpha$  is the coefficient for controlling the yield curve transition. To generate the flow curve, equation 6 is rearranged to find the stress value based on the strain value, which is then plotted to obtain the flow curve as in figure 12. The modified equation, which the values are directly inserted to Abaqus material parameters definition, can be written as follows:

$$\sigma = \sigma_y \left( \frac{\varepsilon}{\alpha \varepsilon_y} \right)^{\frac{1}{n}} \quad (7)$$

Important note that the yield stress is defined as 300 MPa based on table 1, hence the yield strain can be calculated as follows:

$$\varepsilon_y = \frac{\sigma_y}{E} = \frac{300 \text{ MPa}}{210000 \text{ MPa}} = 0.00142857 \quad (8)$$

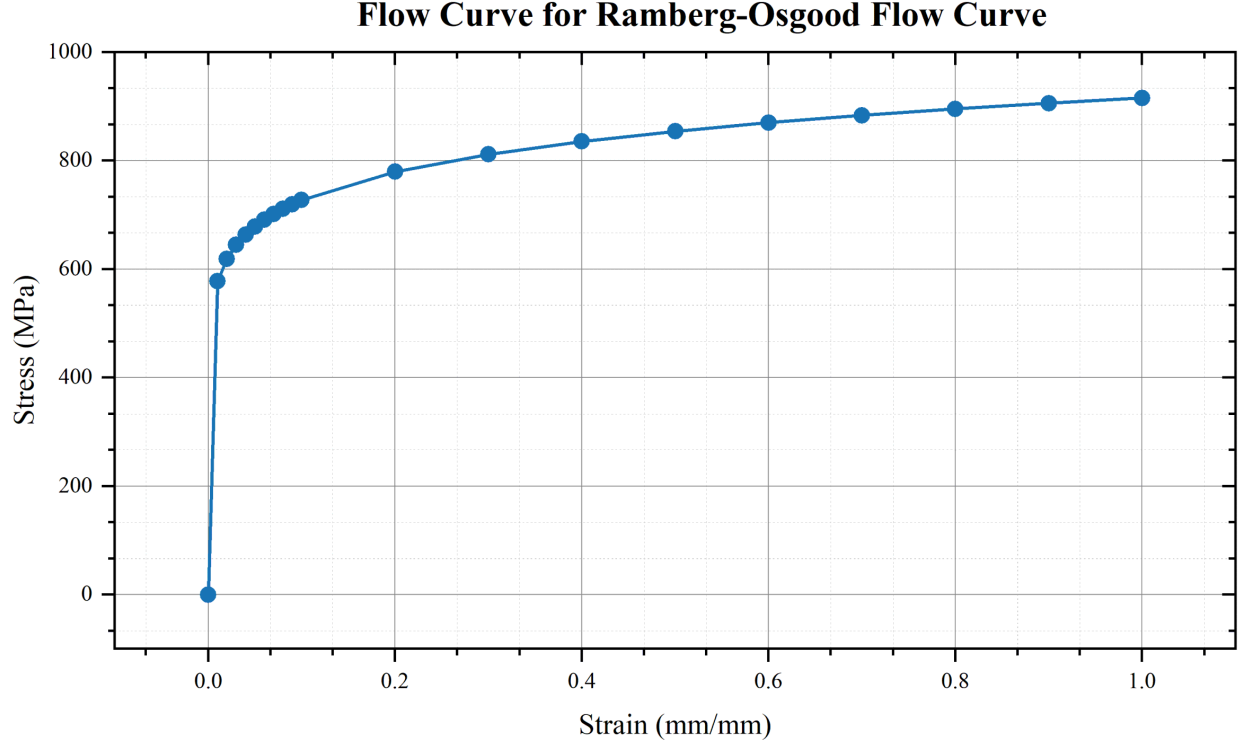


Figure 12: Ramberg-Osgood flow curve based on the Ramberg-Osgood equation with  $\alpha = 0.01$  and  $n = 10$

After the elastic-plastic behavior is defined, the J-integral values are evaluated for both elastic and elastic-plastic cases in plane stress condition. The J-integral values and the von Mises stress distribution results are plotted for elastic and elastic-plastic condition in figure 13. From the figure, it can be observed that J-integral values for both cases show a converging trend as the contour number increases. For elastic case, the representative J integral value can be taken from contour number 5 until 10, which around 5.5 N/mm. Meanwhile, for elastic-plastic case, the representative J integral value can be taken from contour number 5 until 10 as well, which around 2.26 N/mm. For both elastic-plastic and elastic case, the difference can be observed from the discrepancy of the J-integral values, as the elastic-plastic case have smaller values compare to elastic case, which shown in shifting downward of the elastic-plastic curve. This relation also shown within von Mises stress distribution results around the crack tip, where the elastic case shows higher stress concentration compare to the elastic-plastic case.

This discrepancy can be explained through the fact, that J-integral have dependency to the stress field, which for both relation of the J-integral value as well for the stress field are in a good agreement. The higher stress-field results can be explained in figure 7, whereas in the force-displacement controlled test, the elastic case produced a larger force compared to the elastic-plastic case. This is due to the fact, that in the elastic case, larger force will be produced in a higher displacement since softening is not introduced in this model. Therefore, the results showed a good agreement with the physical interpretation.



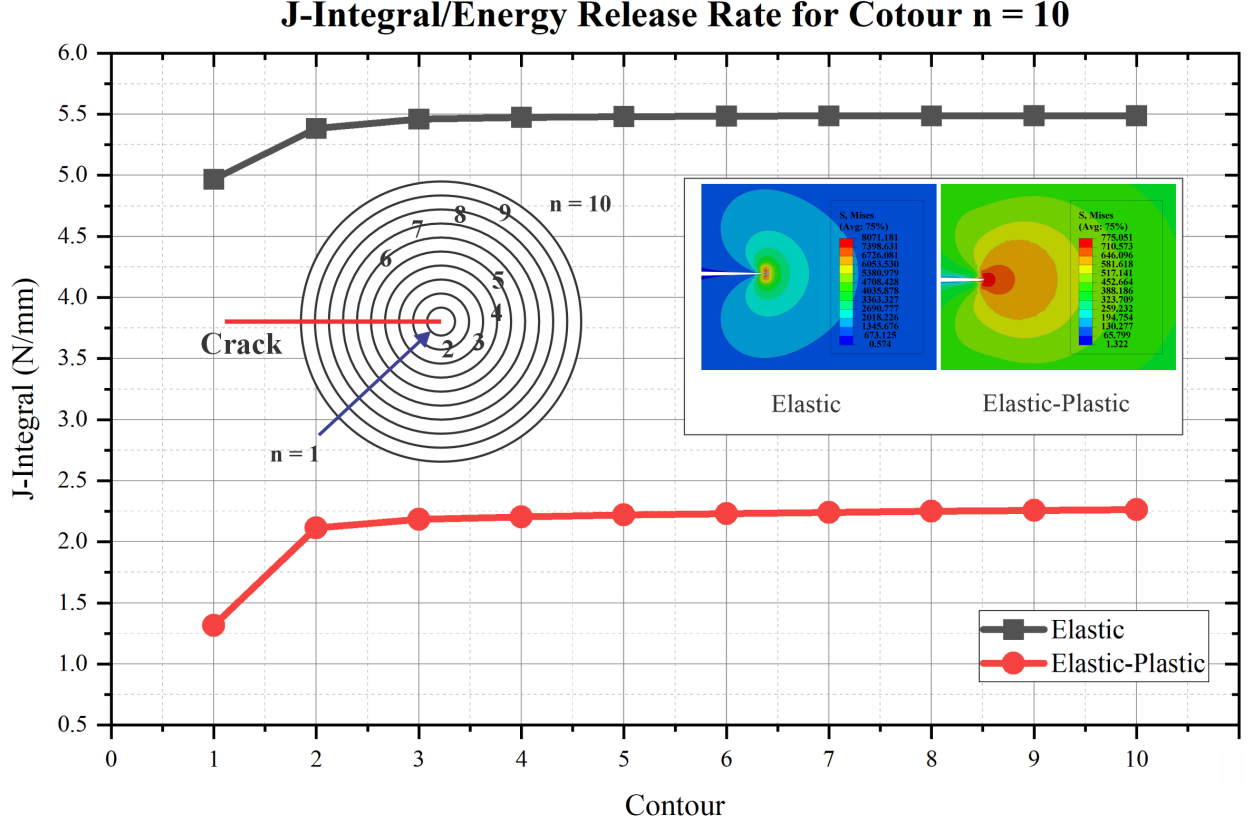


Figure 13: J-integral values over 10 contours around the crack tip for both elastic and elastic-plastic analysis under plane stress condition. The measurement technique used is domain integral method, as the contour integral is evaluated from  $n=1$  until  $n=10$ .

To validate the J-integral results, evaluating the theoretical J-integral value is conducted to determine the FEM accuracy. The first step is by knowing the stress intensity factor  $K_I$  for the crack tip, which can be calculated using the following equation:

$$K_I = \frac{P}{B\sqrt{W}} f\left(\frac{a}{W}\right) \quad (9)$$

Where  $P$  is the applied load,  $B$  is the thickness (considered as 1 mm for plane stress condition),  $W$  is the width of the plate, and  $a$  is the crack length. The function  $f\left(\frac{a}{W}\right)$  is a geometry correction factor, which can be calculated using the following equation:

$$f\left(\frac{a}{W}\right) = \frac{2 + \frac{a}{W}}{\left(1 - \frac{a}{W}\right)^{3/2}} \left( 0.886 + 4.64 \frac{a}{W} - 13.32 \left(\frac{a}{W}\right)^2 + 14.72 \left(\frac{a}{W}\right)^3 - 5.60 \left(\frac{a}{W}\right)^4 \right) \quad (10)$$

This equation is obtained based on literature Zhu et al. (2010) for compact C(T) specimen under mode I loading condition [2]. This is the closest model that can be found, since both set up has similar test configuration, whereas in the literature, it used additional two holes

in the geometry to introduce the concentrated force. Figure 14 shows the C(T) geometry with the geometry dimension nomenclature details.

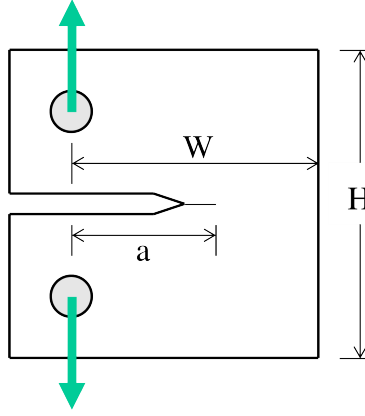


Figure 14: The compact tension (C(T)) specimen based on literature [2]. Two green arrows shows the applied concentration forces to introduce crack propagation.

Another literature [1] mentioned by Clarke, that the geometry factor can be calculated using their proposed equation for C(T) specimen:

$$f\left(\frac{a}{W}\right) = 29.6\left(\frac{a}{W}\right)^{0.5} - 185.5\left(\frac{a}{W}\right)^{1.5} + 655.7\left(\frac{a}{W}\right)^{2.5} - 1017\left(\frac{a}{W}\right)^{3.5} + 638.9\left(\frac{a}{W}\right)^{4.5} \quad (11)$$

Hence, the equation uses different factors to describe the relation of the ratio of the crack length and the specimen width. For  $\frac{a}{W} = \frac{0.504}{2.4} = 0.21$  hence  $f[2] = 4.401$  using the equation 14. While using the equation 11 the results is  $f[1] = 5.217$ .

The last aspect that need to be addressed is the loading P. Whereas both literatures implemented concentrated forces applied in both gauge holes to place the loading, which shown in the green arrows in figure 14. To obtain the equivalent load from the simulation, the reaction force from the applied displacement is extracted from the FEM simulation by probing the reaction force in the y-direction on the left edge of the plate. Based on the probing values in Abaqus, the reaction force is obtained as 205.559 N for the elastic case, while for elastic-plastic case the observed reaction force is 97.5102 N.

After the stress-intensity factor is obtained, the theoretical J-integral value can be calculated. Noted from the literature [2], the J integral is distinguished based on the material behavior. For elastic case, J-integral can be calculated using the following equation:

$$J_{el} = \frac{K_I^2}{E'} \quad (12)$$

Where  $E'$  is the effective Young's modulus, which for plane stress condition is equal to E. Meanwhile, for elastic-plastic case, J-integral can be calculated using the following equation:

$$J = J_{el} + J_{pl} = \frac{K_I^2}{E'} + \frac{\eta_{pl} A_{pl}}{Bb} \quad (13)$$

Where  $\eta_{pl}$  is a geometry factor,  $A_{pl}$  is the plastic area under the load-displacement curve,  $B$  is the thickness, and  $b$  is the uncracked ligament length ( $b = W - a$ ). For the plastic area based on the elastic-plastic force-displacement curve, the calculated area is 0.569 N.mm.

While the geometry factor can be calculated using referenced equation from Xian-Kui [2], which can be described as following equation:

$$\eta_{pl} = 2 + 0.522 \frac{b}{W} \quad (14)$$

A table is created to summarize the theoretical J-integral values calculation for both elastic and elastic-plastic case. The summary also consist of calculation using different geometry factor  $f$  which obtained from different literature as a comparison of both parameterization performance.

Table 3: Theoretical J-integral value calculation summary for elastic and elastic-plastic case

| Material Model  | P (N)   | $K_I[2]$ (MPa $\sqrt{m}$ ) | $K_I[1]$ (MPa $\sqrt{m}$ ) | $J_{el}$ (N/mm) [2] | $J_{el}$ (N/mm) [1] |
|-----------------|---------|----------------------------|----------------------------|---------------------|---------------------|
| Elastic         | 205.559 | 583.959                    | 692.232                    | 1.624               | 2.282               |
| Elastic-Plastic | 97.5102 | 277.010                    | 328.371                    | 0.365               | 0.513               |

Table 4: Final results of the theoretical J-integral value calculation summary for elastic and elastic-plastic case (continue from table 3).

| $A_{pl}$ (N.mm) | $\eta_{pl}$ | $J_{pl}$ (N/mm) | Total J[2] (N/mm) | Total J[1] (N/mm) | Sim. Result (N/mm) |
|-----------------|-------------|-----------------|-------------------|-------------------|--------------------|
| –               | –           | –               | <b>1.624</b>      | <b>2.282</b>      | <b>5.5</b>         |
| 0.569           | 2.412       | 0.724           | <b>1.089</b>      | <b>1.237</b>      | <b>2.26</b>        |

Based on the theoretical calculation from both literatures, all values for each reference are not in a good agreement with the simulation results. However, a good agreement showed in relation of elastic-plastic and elastic case for the calculated results, where the elastic case J-integral is consistently larger than the elastic-plastic case in the load-displacement controlled experiment, according to the relation for J-integral comparison in figure 13. Hence, the discrepancy can be explained from the theoretical point of view, that the limitation came from the parameterized factor  $f(\frac{a}{W})$ , where from both literatures are using the C(T) experimental set up as shown exactly in figure 14. This may come from the force set up, as in the experiment two holes are used as the referenced hinged-force placement. This concludes that the present simulation model is not fully equivalent to the experimental set-up from both literatures.

From the simulation point of view, these discrepancies are likely not cause by the simulation results since the J-integral plot have already shown a converged result. Therefore, a geometry factor specifically for this test-case set up should be used to validate the accuracy of the modelling.

## Q2: Crack propagation using a Cohesive Surface

### Traction-separation law

... Construct the traction-separation curve using the cohesive parameters in Table 2. Calculate the fracture energy from this curve and briefly describe the meaning of each segment.

The first step to construct the traction-separation curve is to determine the known parameters, which has been defined in table 2. Hence, the stiffness values in normal and shear direction are  $K_n = K_s = K_t = 210 \text{ GPa} = 210000 \text{ MPa}$ . The maximum nominal stress in normal direction is  $\sigma_n = 500 \text{ MPa}$ , while the shear direction maximum nominal stress is  $\sigma_{s,t} = 0 \text{ MPa}$ . The failure separation displacement is  $\delta_{fail} = 0.0038 \text{ mm}$ . This information is then used to draw the curve, by plotting the stiffness as the slope of the curve in the initial linear region, until it reaches the maximum nominal stress. After knowing the maximum value, a line is drawn from the maximum nominal stress to the failure separation displacement point (hence the stress value is 0), which represents the softening behavior of the cohesive zone. The final traction-separation curve is shown in figure ??.

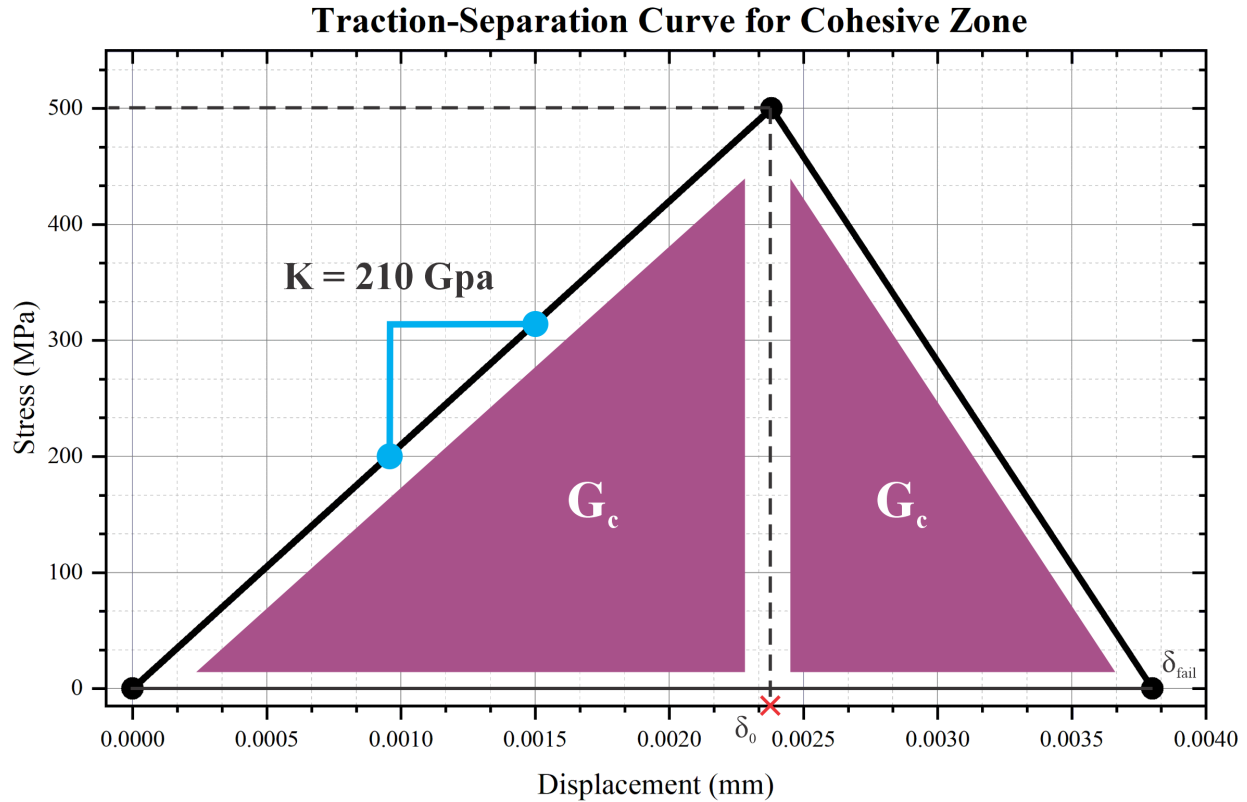


Figure 15: Traction-separation curve based on the cohesive parameters defined in table 2. The curve shows the initial linear elastic region, followed by the softening region until failure separation displacement is reached.

## Crack propagation plots

... Include a figure of the selected mesh and specify the mesh size used in the simulation. Plot the relevant stress components at several load increments to clearly show how the crack initiates and propagates along the cohesive surface.

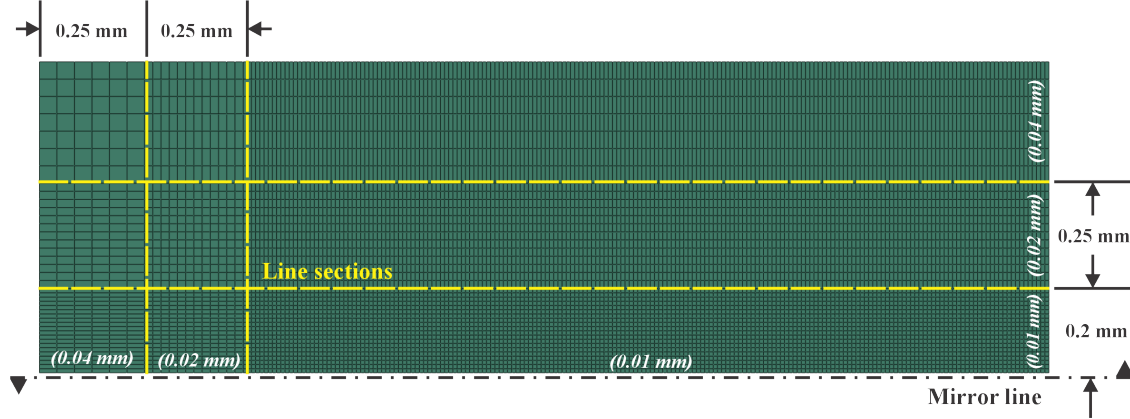


Figure 16: A half-cut mesh configuration for the cohesive zone model in question 2 for the top part geometry. A fine mesh is used around the cohesive area to accurately capture the stress distribution during crack propagation. The mesh size in the cohesive area is 0.01 mm, with an increase mesh transition of 2x of the element size, which at the neighboring elements is 0.02 mm and 0.04 mm respectively. This is used to ensure a smooth transition in the mesh size, while still maintaining accuracy around the cohesive zone and decrease computational cost for the non-prioritized regions.

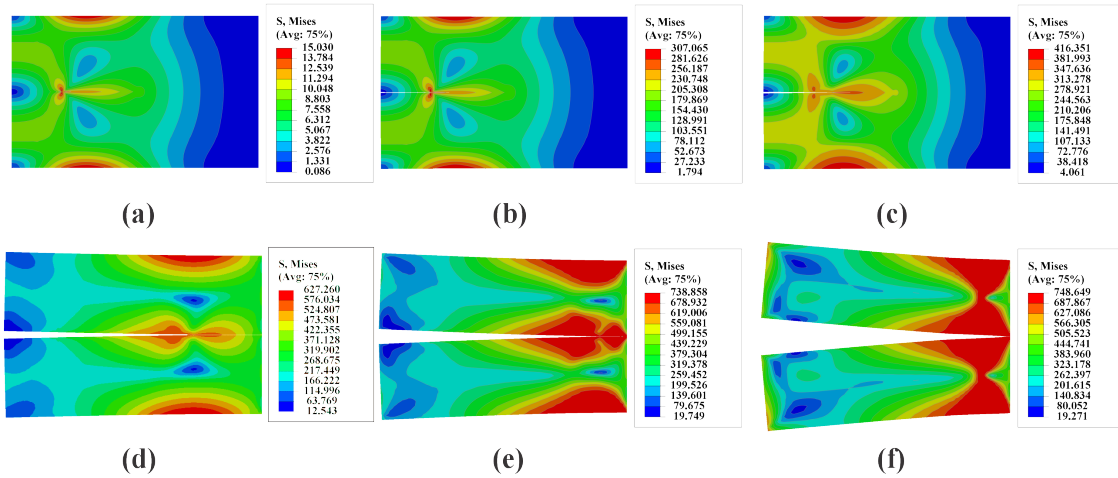


Figure 17: Crack propagation plot showing the von Mises stress distribution at several load increments. The crack initiates and propagates along the cohesive surface through stages (c) to (f). Stage (a) and (b) shows the pre-crack initiation, indicating a constant stress distribution with an increase in its value before the crack initiates.

### Force-time response

... Compute the total reaction force in the y-direction on the left edge of the upper part. Plot force versus time and summarize how the slope changes as the crack initiates and propagates.

To compute the force vs. time response, the reaction force ( $R_{fy}$ ) in each node of the left upper edge (where the displacement is applied) is probed in the y-direction. This will give values of reaction force for every single mesh node, which then all of probed reaction force values are summed to obtain the total reaction force. Abaqus has automatically plotted the total reaction force vs. time response. Therefore, the total force can be calculated using the following equation:

$$R_{fy,total}(t) = \sum_{i=1}^n R_{fy,i}(t) \quad (15)$$

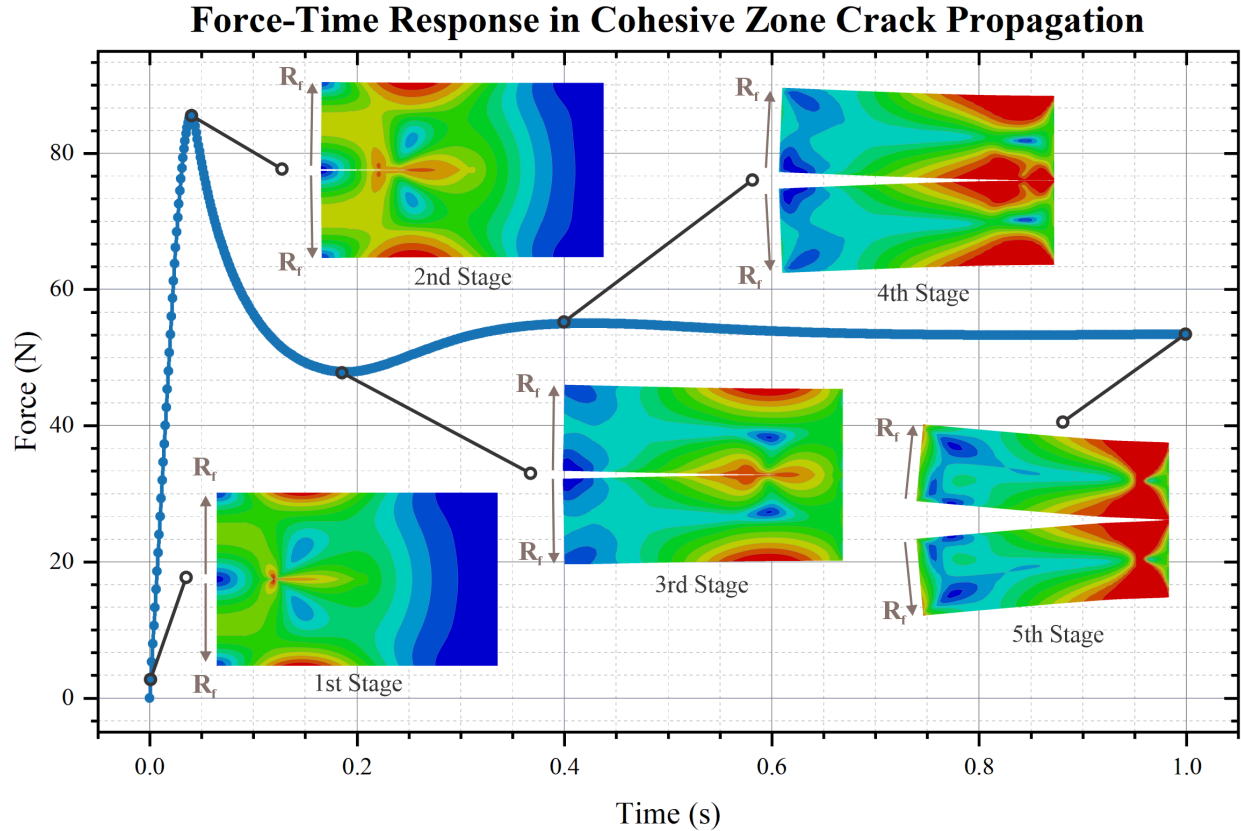


Figure 18: Force-time response showing the total reaction force in the y-direction on the left edge of the upper part. The slope changes indicate crack initiation and propagation for stage 1 until stage 5.

To analyze the force behavior in the cohesive format, figure 18 shows the plotted force-time response for the cohesive zone model in a plane strain condition. 1st stage shows a

constant stress distribution (this also can be referred in figure 17 from (a) to (b)), where the force increases linearly with time. This step is also known as the pre-crack initiation, as the specimen remains in the linear region. As the crack initiates (stage 2), the slope begins to decrease, indicating the initial damage in the cohesive zone. This makes a decrease in the force value. As the crack propagates further, stage 3 shows a minimum drop point of the force value, whereas the crack continues to propagate (stage 4 - stage 5) the values remains relatively constant until the crack opening reach its maximum separation distance.

During the 1st stage until the initial crack initialize, this region is known as a linear elastic region. Refer to the traction-separation curve in figure ??, this region is represented by the initial linear region until it reaches the peak nominal stress that is input in Abaqus user definition, which is 500 MPa. As it reaches the peak nominal stress, the crack initiates and starts to propagate, which is represented by the 2nd stage until the final stage in figure 18. The decrease in the slope force indicates that the cohesive zone is undergoing softening behavior, implied by the decrease of the load-bearing capacity of the cohesive zone as the separation displacement increases. This will reach a minimum point, where the force is at the lowest state, shown is 3rd stage.

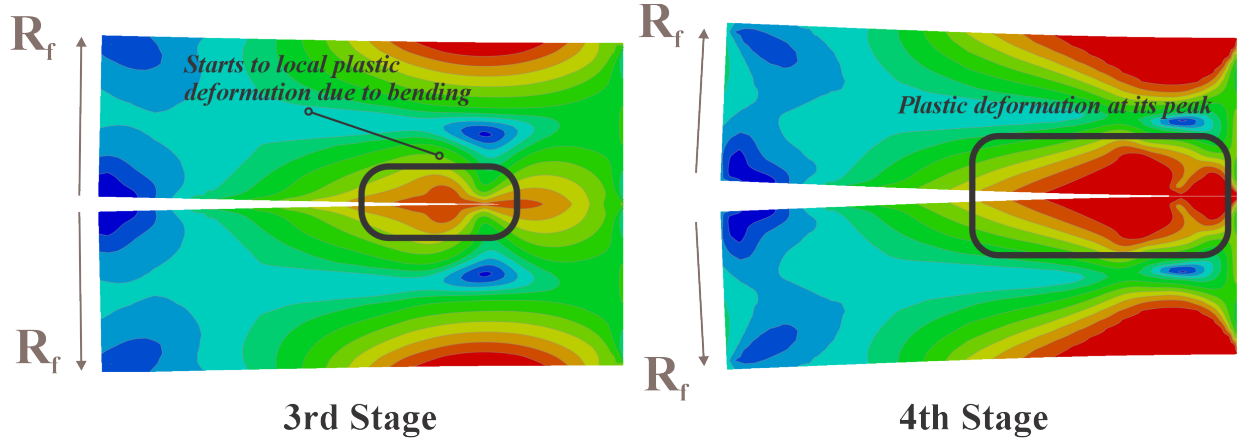


Figure 19: Plastic deformation around the crack tip during the crack propagation. The plastic deformation is observed to occur around the crack tip, indicating that the plate is undergoing plastic bending during the loading.

As the crack propagates further, there is a slight increase in the force value, where it peaks located in the 4th stage. These phenomena can be explained by the fact that the event is not purely contribute from the separation displacement in the normal direction, but also the plate plasticity plays a role during the deformation. As shown in preceeding figures, since the plate has quite high aspect ratio ( $b/h = 1.5625$ ), therefore the plate will have tendency to undergo bending during the loading. Therefore, an increase in the force value is observed since the plastic-bending deformation also contribute to the overall reaction force. After this local bending, the force is relatively constant until the crack reaches its maximum separation displacement.



## Cohesive Surface Analysis

... Extract traction values at three selected points along the cohesive surface. Compare these results with the analytical traction-separation curve and provide a concise conclusion.

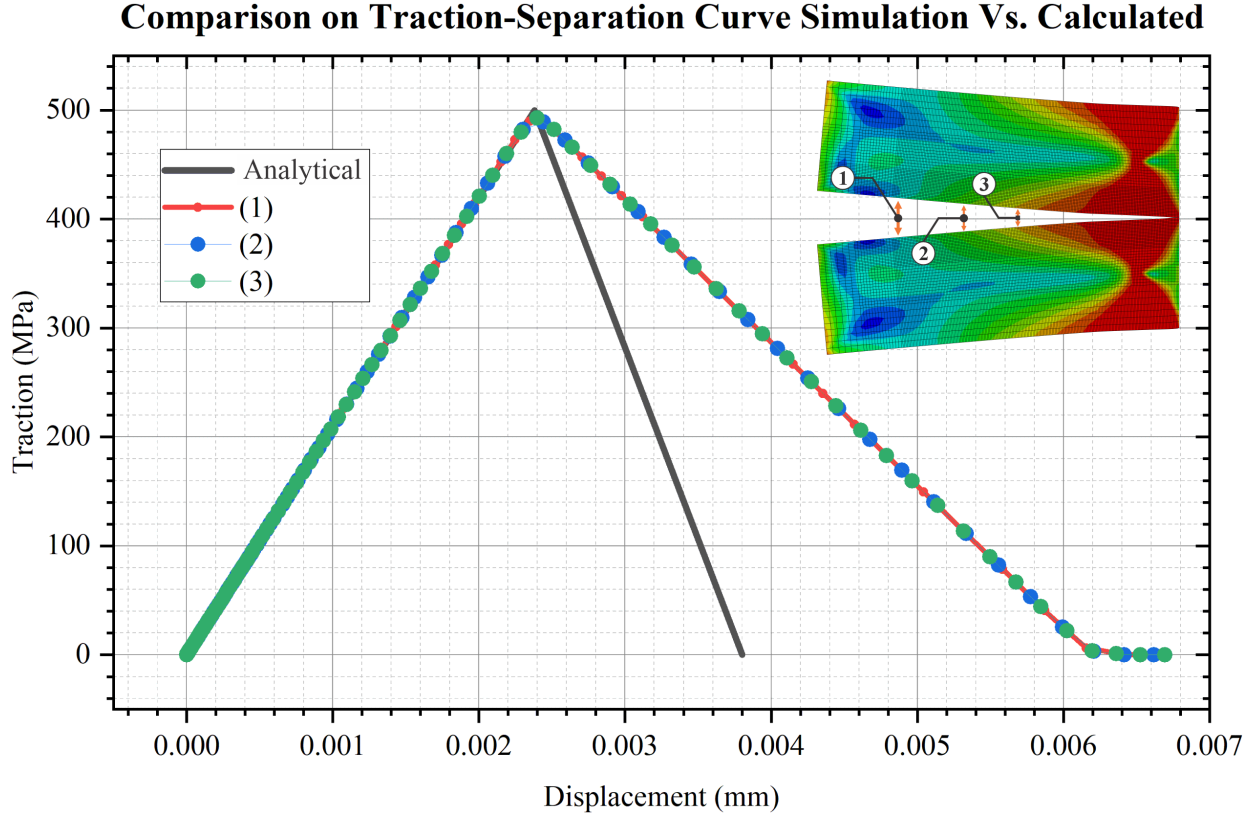


Figure 20: Comparison of traction values at three selected points along the cohesive surface with the analytical traction-separation curve.

To compare the implemented traction-separation curve and the simulation results, figure 20 shows both plots, showing the probed location on each of the point location in the geometry. As in the first region, both analytical and simulation result shows the same behavior in the linear increase of the traction. As it reached its peak, which have stress of 500 MPa, the traction starts to decrease in the same starting point of the displacement as well. However, differences start to be observed in the softening part, as the simulated result are having the softer behavior compare to the defined traction-separation curve.

Referring to the implemented traction-separation curve in figure 15, the stiffness is defined for every axis direction, which are in normal and both two shear direction stresses. While, the separation criterion is only implemented for the normal direction only. This softening in the simulated result can be explained by the fact that the stiffness in two shear direction are still exists. Since there is no separation criterion in the stiffness, these shear-stiffness helped the separation process degrade softer instead of in a steep decrease sense as



in the defined curved. Because in the geometry, not only normal stress but shear stresses in two directions are also take part as the main contributor of the stress generation. Therefore, this softening are shown and the separation displacement criterion is increase as well because it takes longer displacement to reach zero traction as the curve is more flattened compare to the steep one.

## Conclusion

To sum up everything that have been analyzed and gathered in this study, here are some several key takeaways from this small case study:

- The higher value of Von-Misses stress in plane stress compare to plane strain is caused by the non-restrictive out-plane strain, which give more plane stress to generate Stresses. While plane strain restrictive out-plane strain will increase the hydrostatic stress, contradictively by knowing the relation of hydrostatic and von-Misses stress, this will reduce the von-Misses stress value.
- In contrast to the Von-Misses value, the generated force in plane strain is larger due to the out-plane strain restriction, since the hydrostatic have relation as well to the  $\sigma_{22}$  component, meaning that the reaction force in the same direction will also increase by this means.
- J-Integral calculation should use a suitable geometry factor parameterization, specifically for the exact boundary condition set up for both simulation and experimental in order to generate a good agreement results comparison of the theoretical and simulation results.
- Softening effect will present in the cohesive zone during separation due to the contribution of ...

## References

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